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RESEARCH ARTICLE

Tracking amyotrophic lateral sclerosis disease progression using passively collected smartphone sensor data

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Abstract

Background: Passively collected smartphone sensor data provide an opportunity to study physical activity and mobility unobtrusively over long periods of time and may enable disease monitoring in people with amyotrophic lateral sclerosis (PALS). **Methods:** We enrolled 63 PALS who used Beiwe mobile application that collected their smartphone accelerometer and GPS data and administered the self-entry ALS Functional Rating Scale-Revised (ALSFRS-RSE) survey. We identified individual steps from accelerometer data and used the Activity Index to summarize activity at the minute level. Walking, Activity Index, and GPS outcomes were then aggregated into day-level measures. We used linear mixed effect models (LMMs) to estimate baseline and monthly change for ALSFRS-RSE scores (total score, subscores Q1–3, Q4–6, Q7–9, Q10–12) and smartphone sensor data measures, as well as the associations between them. **Findings**: The analytic sample (N = 45) was 64.4% male with a mean age of 60.1 years. The mean observation period was 292.3 days. The ALSFRS-RSE total score baseline mean was 35.8 and had a monthly rate of decline of -0.48 (p-value <0.001). We observed statistically significant change over time and association with ALSFRS-RSE total score for four smartphone sensor data-derived measures: walking cadence from top 1 min and log-transformed step count, step count from top 1 min, and Activity Index from top 1 min. **Interpretation**: Smartphone sensors can unobtrusively track physical changes in PALS, potentially aiding disease monitoring and future research.

Background

Amyotrophic lateral sclerosis (ALS) is an uncommon (incidence of approximately 1–2/100,000) adult-onset fatal neurodegenerative disease that primarily affects motor neurons.[1,2] Median onset is between the ages of 50 and 65, and the most common symptoms are muscle twitches, cramps, and weakness.[1] The pathobiology of ALS is incompletely understood, and the location of onset, pattern and speed of progression, and degree of upper motor neuron involvement vary between patients.[3–5] This heterogeneity contributes to delays in ALS diagnosis and inefficient ALS trial design.[1]

Digital biomarkers can be defined as characteristics, or sets of characteristics, collected from digital health technologies and measured as indicators of normal biological processes, pathogenic processes, or responses to exposure or therapeutic interventions.[6] Smartphones, one type of digital health technology, provide a unique opportunity for digital biomarker data collection. For one, smartphones have become nearly omnipresent; over 85% of the United States population owns a smartphone.[7] In comparison, clinical-grade wearable devices can be cost-prohibitive and may result in low adherence, as patients are not accustomed to donning wearable devices near-continuously. Additionally, smartphones provide opportunities for the collection of both active data (surveys, voice recordings) via mobile apps and passive data (accelerometer, gyroscope data) via smartphone sensors.[8]

To date, therapeutic development for ALS has relied heavily on staff-administered functional rating scales to measure ALS progression. The Revised Amyotrophic Lateral Sclerosis Functional Rating Scale (ALSFRS-R) is the primary endpoint in the vast majority of contemporary ALS trials.[9–12] The ALSFRS-R is a staff-administered scale consisting of 12 items; each item is given a score of 0–4 points, with a higher score indicating better performance or function. While it is a very useful scale, it does not meet unidimensionality criteria,[10,13] it is ordinal (i.e., point-wise decline is nonuniform with each score decrease), it may fail to detect decline in people with ALS (PALS) during a standard trial period of 6 months,[12] and it requires staff administration and training.

The self-entry ALS Functional Rating Scale-Revised (ALSFRS-RSE) overcomes some of the ALSFRS-R limitations and has repeatedly demonstrated excellent correlation with the ALSFRS-R.[14] A key advantage is that it can facilitate home-based monitoring of disease progression thereby reducing the burden of frequent clinical visits. Additional self-entry scales exist, including the Rasch-built Overall ALS Disability Scale (ROADS).[13] Research has confirmed the feasibility of ALS patient-reported outcomes both in-clinic and out-of-clinic.[15,16]

Remote, passive data collection is another novel approach that can address key limitations of traditional rating scales like the ALSFRS-R, expand data robustness, and reduce patient burden. Wearable devices and smartphones allow for objective, detailed, and near-continuous measurements of human activity over long periods of time.[17] The use of digital biomarkers is increasing across medicine, and the viability of using devices and sensors for continuous patient data collection in the free-living environment, rather than in the clinic, has been demonstrated for a number of measurements: heart rate variability, accelerometry-based activity, speech, spirometry, hand-grip strength, and electrical impedance myography.[18–21] Several studies have assessed continuous monitoring of PALS using wearable devices; these studies have shown that accelerometer-based endpoints of physical activity (PA) levels are strongly correlated with the ALSFRS-R total score[14,18,19–21] and the gross motor domain score.[20]

Smartphone apps offer novel opportunities for remote and unobtrusive behavior monitoring over extended periods via both active and passive data collection without requiring the implementation of a wearable device. In a previous study by Panda et al.,[22] smartphone accelerometer data were used to quantify PA volume in patients after cancer surgery. To the best of our knowledge, neither smartphone accelerometer nor GPS data-based measures have previously been used to study PA volume, walking, or mobility in PALS.

A potential challenge in remote, passive data collection is data missingness, identified as a common issue in digital health studies.[23,24] In the context of smartphone-based monitoring, Kiang et al. (2021) considered six different studies and reported data missingness to be 19% for accelerometer data and 27% for GPS data.[25] The authors argued that the sensor noncollection could be due to factors such as user behavior (not charging the phone, turning off GPS, or removing the study app) and technological constraints (like limitations set by the phone's operating system on data collection). For accelerometer data, they found that "Black participants had higher rates of non-collection, but rates did not differ by sex, education, or age." For GPS data, they found that "iOS users had lower GPS non-collection than Android users", but "non-collection did not differ by race/ethnicity, education, age, or gender."

Our aim was to explore the potential of using a smartphone mobile application (app) for functional assessment in PALS. In the present study, we continuously monitored 45 PALS for up to 1 year using smartphones for both active and passive data collection. Our aims were to (1) use raw tri-axial smartphone accelerometer and GPS data to derive day-level measures of walking, Activity Index, and mobility; (2) estimate the baseline levels and monthly change over time of walking, Activity Index, and mobility measures; and (3) quantify the association between measures of walking, Activity Index, and mobility measures with ALSFRS-RSE total score and ALSFRS-RSE subscores corresponding to bulbar, fine motor, gross motor, and respiratory function.

Methods

Study ethics

Approvals from the Institutional Review Board (IRB) at Massachusetts General Brigham (MGB) and the Information Security Office at Massachusetts General Hospital (MGH) were obtained before study initiation. All data collection and management adhered to MGB, state, and national guidelines and regulations. Participants provided informed consent prior to engaging in study procedures.

Study design and population

This study was conducted at a single center and was noninterventional. Eligible participants were ambulatory at enrollment, at least 18 years of age, and diagnosed with ALS according to the El Escorial Criteria[26], and able to provide informed consent, follow study procedures, and use their smartphone independently, as assessed by the study investigator. The study was advertised on ALS social media accounts, on an institutional research recruitment website, and at the ALS multidisciplinary clinic using IRB-approved materials. Participants were enrolled, and study procedures were conducted remotely, except in rare cases (participant on site).

Data collection

Demographic data were collected at baseline via phone/television interviews. Remote self-entry ALSFRS-R (ALSFRS-RSE), a standardized clinical outcome assessment questionnaire, and smartphone sensor data (accelerometer, GPS) were obtained using the Beiwe smartphone application (app). The Beiwe app serves as the front-end of the open-source Beiwe high-throughput digital phenotyping platform,[27] which includes apps for Android and iOS, a web-based platform for study administration, and a cloud-based system for data storage and processing. All procedures for Beiwe data collection and storage were carried out in compliance with local, state, and national regulations. Participants were instructed on how to download and use the Beiwe app on their phones (Android or iOS) during the baseline visit procedures. Participants contributed smartphone data for up to 1 year. The Beiwe app was removed upon completion of the study. Participants were able to contact study coordinators for technical assistance, though these calls were not systematically logged.

The Beiwe app was configured to administer ALSFRS-RSE at baseline and every 2–4 weeks thereafter and to collect accelerometer and GPS data from participants' smartphones for the duration of the study. Modern accelerometers are small electro-mechanical devices that record phone acceleration along three orthogonal axes at high frequency.[17] Beiwe was configured to collect raw (unprocessed) accelerometer data at a sampling frequency of 10 Hz in 10-sec on- and 10-sec off-cycles. GPS data were collected in 1-min on- and 10-min off-cycles to minimize battery drain and optimize data storage[28] (see Supplementary Material: Beiwe configuration file).

Processing of smartphone raw accelerometer data

Accelerometer data were preprocessed to derive minute-level physical activity measures, including walking characteristics (step count, average cadence expressed in steps per second) and a nonnormalized minute-level Activity Index[29] measure. The Activity Index measure aggregates raw accelerometer data into a single value per minute, with higher values corresponding to higher intensity of a movement recorded with an accelerometer. For detailed information on the processing of smartphone raw accelerometer data, refer to the Supplementary Material.[30,31]

We derived six day-level (24-h data collection period) measures by aggregating minute-level data for walking and Activity Index. These measures are log(Activity Index sum from all day minutes), log(Activity Index from top 1 min), walking cadence from all day minutes (step count-weighted average), walking cadence from top 1 min, log(step count from all day minutes), and log(step count from top 1 min). The term "top 1 minute" refers to the minute with the highest recorded value for a specific measure on a given day and is consistently used throughout the manuscript. The term "from all day minutes" is further omitted for simplicity in the manuscript. The term "log()" denotes a natural logarithm-transformed quantity +1 for the quantity specified in the brackets ("+1" was used to cap the lower range of the outcome at 0 and is omitted in the notation).

We also defined a "valid" minute flag and a "phone on person" minute flag for accelerometer data. In brief, the first flag determines whether the accelerometer sensor was properly collecting the data; the latter indicates whether the participant's phone was in motion (e.g., held in hand or in a pocket) or perfectly still. See details in Supplementary Material: Processing of smartphone raw accelerometer data.

Processing of smartphone GPS data

In this study, we used GPS data collected in a 1-min on-cycle followed by a 10-min off-cycle, which resulted in missing data by design. As demonstrated in previous work,[32] such missing GPS data must be imputed prior to constructing day-level summary statistics. To achieve this, we employed our established GPS imputation method.[32,33] This approach converts GPS data into a sequence of flights (segments with movement) and pauses (segments with no movement) and imputes missing segments using the empirical distribution estimated from the observed data using resampling weights that depend on location and time of day. We used our second-generation version of the method which retains the original spherical geometry of the problem, imputes gaps bidirectionally, and makes use of sparse online Gaussian processes for efficient and accurate imputation.[33] The method is included in our Forest library for Beiwe data processing on GitHub (https://github.com/onnela-lab/forest). After imputation, we calculated two continuous GPS-based day-level (i.e., 24 h level) measures: log(distance traveled) (in kilometers) and time spent at home (in hours).

Analytic sample

We defined a "valid day" of smartphone accelerometer data for a given participant as a day in which the number of valid minutes falls within the mean of the participant's daily valid minutes ±180 min band and if the day falls within a 28-day window with at least seven such days. For GPS data, we included a day in the GPS data analysis if there were any GPS data collected on that day and if it fell within a 28-day window with at least seven such days (for details and elaborated reasoning for those decisions, see Supplementary Material: Analytic sample; for quantification of stability of a valid minutes pattern in the analytic sample, see Methods: Statistical data analysis).

For this study, we analyzed data from participants who fully completed at least two ALSFRS-RSE surveys and had at least 28 days of valid accelerometer data and at least 28 days of valid GPS data. The participant's observation period was defined as the timeframe between the start of study participation and either the completion of a 1-year period, the participant's withdrawal or death, or the calendar date of September 6, 2022 (data pull from the database), whichever came earliest.

Statistical data analysis

We characterized smartphone survey frequency and sensor data quality by calculating several metrics for each participant. These metrics included the number of ALSFRS-RSE survey submissions, the average number of days between submissions, the number of days with valid accelerometer and GPS data included in the analysis, and the average number of valid accelerometer minutes per day during the whole observation period and in for subset of days included in the analysis. We then aggregated (median and range) these statistics across participants.

We used linear mixed effect models (LMMs) to analyze the ALSFRS-RSE scores and day-level measures based on smartphone sensor data. To quantify the average ALSFRS-RSE at baseline and its monthly change, five LMMs were fit. Each of these models specified time as a fixed effect and included a participant-specific random intercept and random slope but had different outcomes: ALSFRS-RSE total score, ALSFRS-RSE questions 1–3 subscore (Q1–3; bulbar function), Q4–6 (fine motor function), Q7–9 (gross motor function), and Q10–12 (respiratory function).

We employed LMMs to quantify the stability of the daily number of valid minutes and "phone on person" minutes over time. Two separate LMMs were fitted, with daily statistics as the outcome variable, time as a fixed effect, and participant-specific random intercept and random slope included.

We derived six day-level measures from smartphone accelerometer data and two day-level measures from smartphone GPS data. To quantify baseline and monthly change for the day-level measures, we fit eight LMMs, one for each measure. Each model included one day-level measure as an outcome, time as a fixed effect and participant-specific random intercept and random slope.

To assess the association between the measures based on smartphone sensor data and ALSFRS-RSE outcomes, we fitted LMMs to each pair of one of the eight measures and one of the 5 ALSFRS-RSE scores (total score, Q1–3, Q4–6, Q7–9, Q10–12). Each model included ALSFRS-RSE score as a fixed effect and had participant-specific random intercept and random slope. The outcome was constructed by averaging the day-level measure's values spanning 21 days (10 days before and after the survey date for a given participant and survey). Given that mobility and physical activity levels are typically associated with day of the week, we intentionally chose a multiple of 7 (here, 21) to ensure that each day of the week is equally weighted in the average.

In the models described above, the fixed effect of time was expressed in months elapsed since a participant's study started. For statistical inference, we employed the Benjamini-Hochberg (BH) correction for multiplicity testing to control a false discovery rate (FDR) at 0.2 for all 55 effects examined. We present the original p-values and designate an effect as statistically significant by denoting it with an asterisk whenever the BH-adjusted p-value is less than 0.2. Wherever applicable, we report standard deviation of the random residual error, standard deviation of the random slope, and the conditional coefficient of determination (R2c) for LMMs.[34,35]

We performed all analyses using R software (version 4.1.2; The R Project). The R code for data preprocessing and analysis is available on GitHub (https://github.com/onnela-lab/als-beiwe-passive-data).

Results

Sample and data quality characteristics

Data from 63 PALS enrolled between January 2021 and July 2022 were considered in this study. Out of 63, a subset of 54 PALS met the condition for having at least two ALSFRS-RSE surveys completed (57 had any), 52 had at least 28 days of valid accelerometer data, and 53 had at least 28 days of valid GPS data. Taking intersection of these subsets, the analytic sample consisted of 45 PALS. The analytic sample was 64.4% male, with a mean age of 60.1 years (Table 1). Overall, there was a high proportion of white participants (91.1%), and most participants' smartphones ran iOS (82.2%). At baseline, the median ALSFRS-RSE score was 37. Median time since symptom onset (in months) was 50 and majority of patients had nonbulbar symptom onset (31 vs. 8 out of 39 participants subset for whom the onset was recorded). Compared with the analytic sample (n = 45), the PALS excluded from the analysis (n = 18) had a slightly lower proportion of males (64.4%), higher mean age (63.2 years), a similar proportion of white participants (88.9%), and lower median baseline ALSFRS-RSE score (32.5; calculated for 12 out of 18 PALS for whom it was available).

The mean observation period per participant was 292.3 days, with a median of seven ALSFRS-RSE surveys submitted over this time period (Table 2). The mean number of accelerometer data days included in the analysis was 113.7 (SD = 58.9), with an average number of accelerometer valid minutes per day of 724.7 (SD = 158.0) (maximum possible: 1440 valid daily accelerometer minutes; see Fig. 1). The mean number of GPS data days included in the analysis was 217.4 (SD = 86.4), with an average number of GPS collection minutes per day of 87.7 (SD = 22.7) (see Figs. S1 and S2 for participant-level characteristics of smartphone accelerometer and GPS data days included in the analysis).

Table 1. Baseline demographic and clinical characteristics of the analytic sample (n = 45) at baseline.

Characteristics	Value
Age	
Mean (SD)	60.1 (10.7)
Median [min, max]	62 [34, 81]
Sex	
Male (n (%))	29 (64.4%)
Female (n (%))	16 (35.6%)
Ethnicity	
Not Hispanic or Latino (n (%))	41 (91.1%)
Hispanic or Latino (n (%))	3 (6.7%)
Unknown/not reported (n (%))	1 (2.2%)
Race	
White (n (%))	41 (91.1%)
More than one race (n (%))	2 (4.4%)
Native Hawaiian or Other Pacific Islander (n (%))	1 (2.2%)
Other (n (%))	1 (2.2%)
Phone operating system	
iOS (n (%))	37 (82.2%)
Android (n (%))	8 (17.8%)
Smartphone self-entry ALSFRS-R (ALSFRS-RSE)	
Mean (SD)	36.0 (6.2)
Median [min, max]	37 [19, 48]
Time since symptom onset (months)	
Mean (SD)	62.3 (60.8)
Median [min, max]	50 [3, 281]
Unknown/not reported (n (%))	6 (13.3%)
Symptom onset site	
Nonbulbar (n (%))	31 (68.9%)
Bulbar (n (%))	8 (17.8%)
Unknown/not reported (n (%))	6 (13.3%)

ALSFRS-RSE, Amyotrophic Lateral Sclerosis Functional Rating Scale-Revised; max, maximum; min, minimum; SD, standard deviation.

ALSFRS-RSE survey baseline and monthly change

Throughout the manuscript, we report original p-values and indicate statistical significance with an asterisk when the Benjamini-Hochberg adjusted p-value is less than 0.2. The LMM-estimated baseline mean for ALSFRS-RSE total score was 35.8 (observed sample mean was 36), with a monthly decline of −0.481 (*p-value <0.001) (Table S1). The majority of participants displayed a downward trend over the study period (Fig. S3). Significant declines were observed in bulbar (ALSFRS-RSE subscore Q1-3; monthly decline est. = −0.112, *p-value <0.001), fine motor (Q4-6; est. = −0.136, *p-value <0.001), gross motor (Q7-9; est. = −0.124, *p-value <0.001), and respiratory (Q10-12; est. = −0.086, *p-value =0.002) functions, indicating progressive deterioration across multiple functional domains in the sample (Table S1). Notably, there was substantial variability in monthly change across participants for ALSFRS-RSE, indicated by large random slope standard deviation (0.424) relative to the fixed effect slope estimate (absolute value = 0.481), and even higher relative variability for the ALSFRS-RSE subscores. With that, the LMM conditional coefficient of determination (R2c) was 0.95 or above across the total score and subscore models, indicating an excellent fit to the data.

Day-level accelerometer data valid minutes, "phone on person" minutes, and monthly change

The average baseline for accelerometer data valid minutes per day, estimated with LMM, was 728.2, with a monthly change of −1.01 (p-value = 0.328). The average baseline for "phone on person" minutes per day was 191.4, with a monthly change of −0.568 (p-value = 0.803). These results show no statistically significant change over time for both accelerometer data valid minutes per day and "phone on person" minutes per day (Table S2) in the analytic sample. In both cases, the monthly change coefficient estimate is also relatively small, accounting for 0.1% and 0.3% of baseline coefficient estimate, respectively.

Table 2. Characteristics of smartphone self-entry ALSFRS-R (ALSFRS-RSE) survey submissions and smartphone data quality in the analytic sample (n = 45). Participant-specific values are aggregated (mean (SD), median [min, max]) across participants.

Characteristic	Values Mean (SD), median [min, max]
Observation period	
# days	292.3 (87.4), 291 [53, 398]
Smartphone self-entry ALSFRS-R (ALSFRS-RSE)	
# survey submissions	7.5 (3.7), 7 [2, 16]
Avg. # days difference between survey submissions	40.9 (25.0), 33 [4, 121]
Smartphone raw accelerometer data quality	
# accelerometer data days included in the analysis	113.7 (58.9), 108 [29, 267]
Avg. # accelerometer valid minutes per day in the observation period	558.7 (195.4), 577 [203, 944]
Avg. # accelerometer valid minutes per day in the analysis	724.7 (158.0), 743 [220, 972]
Smartphone raw GPS data quality	
# GPS valid data days included in the analysis	217.4 (86.4), 212 [52, 381]
Avg. # GPS collection minutes per day in the observation period	68.6 (29.5), 68 [16, 145]
Avg. # GPS collection minutes per day in the analysis	87.7 (22.7), 83 [56, 153]

#, number of; acc., accelerometer; ALSFRS-R, Amyotrophic Lateral Sclerosis Functional Rating Scale-Revised; avg., average; min, minimum; max, maximum; SD, standard deviation.

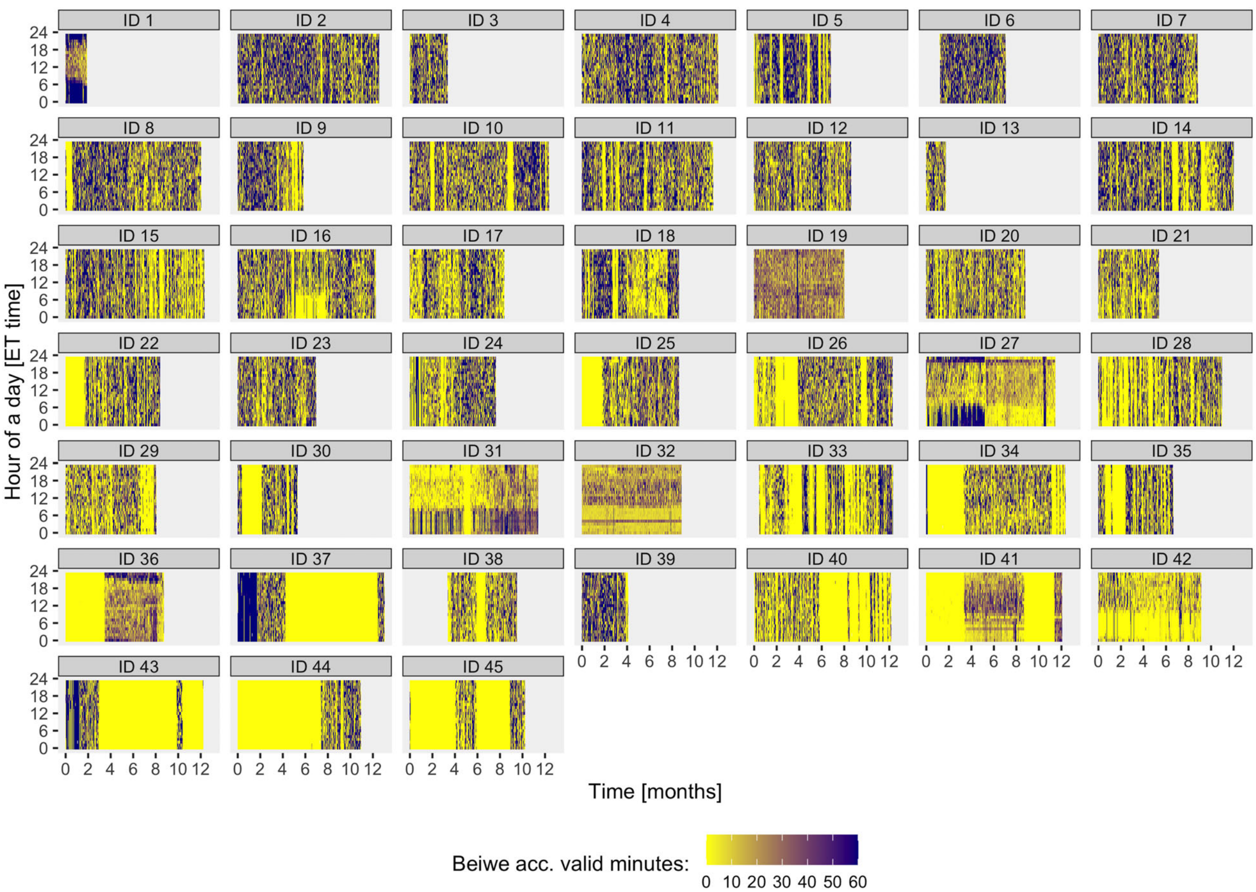

Figure 1. Characteristics of smartphone accelerometer data for each participant. Valid data minutes per hour are shown (color-coded) across hours of a day (y-axis) for each day of participant's monitoring period relative to the start date of monitoring (x-axis). An accelerometer data minute was labeled as valid if it had at least 20 sec (out of 30 sec expected) with at least 5 raw accelerometer data observations each (out of 10 expected).

Hence, although the absence of statistically significant changes over time does not guarantee the absence of real changes, we assumed here it would suffice to plausibly attribute any significant changes in smartphone sensor data-derived measures to underlying changes in waking and mobility rather than changes in smartphone usage duration.

Day-level measures of smartphone sensor data and monthly change

We quantified the average baseline and monthly change of day-level measures of smartphone sensor data. Estimates are shown (Table 3) for eight measures, among which four demonstrated statistically significant monthly decline in function: walking cadence (steps/s) from top 1 min (monthly change est. = 0.004, **p*-value = 0.084) and the log-transformed measures: Activity Index from top 1 min (est. = −0.015, **p*-value = 0.041), step count (est. = −0.090, **p*-value = 0.020), and step count from top 1 min (est. = 0.056, **p*-value = 0.043). The point estimates of average monthly change for all measures reflect decreasing physical activity, as the values for all measures directly related to physical activity were negative. The estimates of average monthly change for GPS-derived mobility metrics (time spent at home, distance traveled) were not statistically significant; directionally, the estimates corresponded to more time spent at home and less distance traveled. We also observed that the random slope standard deviation was large relative to the fixed effect slope absolute value (ratio above 2 for all measures), indicating that the change over time varied considerably across participants. The R2c was less than 0.5 in models for three measures, including walking cadence, walking cadence from top 1 min, log(distance traveled). This suggests that, at least for some of the participants, the change in these measures may not conform closely to a linear trend. (We explored alternative models to fit the data, but the results did not substantially improve in terms of R2c.)

Within 27 days of the participant's first ALSFRS-RSE, Step count, Activity Index, and distance traveled were typically higher for participants with higher ALSFRS-RSE total scores on the first survey, though not all measures followed this trend (Fig. 2). The mean of the median step count from top 1 min from 0 to 27 days after each participant's first ALSFRS-RSE survey across all participants was 20.24 (SD = 20.37), and the mean of the median Activity Index from top 1 min was 19.88 (SD = 21.05) (Table S3).

Association between day-level measures of smartphone sensor data and ALSFRS-RSE

ALSFRS-RSE total score was significantly associated with 6 out of 8 day-level outcome measures (Table 4; participant-level data and population-level trends also visualized in Fig. 3). Time spent at home (in hours) was negatively associated with the ALSFRS-RSE total score (slope est. = −0.076, **p*-value = 0.015), indicating that higher total scores are associated with decreased home time. Log-transformed distance traveled (in kilometers) showed a positive association (slope est. = 0.046, **p*-value <0.001), suggesting that higher total scores are associated with greater distances traveled. Similarly, walking cadence (both overall and from top 1 min) and log-transformed step count (both total and from top 1 min) exhibited significant positive associations with the ALSFRS-RSE total score. We also observed that the random slope standard deviation relative to the fixed effect slope absolute value was below 1 except from one measure (log(Activity Index from top 1 min)). Notably, all models had between good to excellent R2c, ranging from 0.47 (walking cadence from top 1 min) to 0.94 (log(Activity Index from top 1 min)).

Table 3. Average baseline and monthly change (per 1 month) of day-level outcomes derived from smartphone accelerometer and GPS data.

No.	Outcome (y)	Baseline est. [95% CI]	Monthly change est. [95% CI] (*p*-val.)	Resid. SD	Slope SD	R2c
1	Home time (h)	21.25 [20.60, 21.89]	0.028 [−0.039, 0.096] (0.401)	2.952	0.191	0.320
2	log(Distance traveled, km)	2.061 [1.761, 2.362]	−0.024 [−0.054, 0.006] (0.114)	1.382	0.086	0.337
3	log(Activity Index)	6.154 [5.840, 6.469]	−0.016 [−0.042, 0.009] (0.206)	0.601	0.076	0.762
4	log(Activity Index from top 1 min)	2.618 [2.392, 2.843]	−0.015 [−0.029, −0.001] (*0.041)	0.370	0.042	0.827
5	Walking cadence (steps/s)	1.746 [1.723, 1.769]	−0.001 [−0.005, 0.002] (0.364)	0.110	0.007	0.188
6	Walking cadence (steps/s) from top 1 min	1.885 [1.859, 1.912]	−0.004 [−0.008, 0.001] (*0.084)	0.146	0.009	0.168
7	log(Step count)	3.460 [2.659, 4.261]	−0.090 [−0.165, −0.015] (*0.020)	1.914	0.224	0.613
8	log(Step count from top 1 min)	2.496 [1.968, 3.025]	−0.056 [−0.111, −0.002] (*0.043)	1.435	0.162	0.554

CI, confidence interval; Est., coefficient estimate; log, natural logarithm transformation; *p*-val., *p*-value associated with a coefficient estimate (an asterisk denotes a BH-adjusted *p*-value being less than 0.2); R2c, LMM conditional coefficient of determination; resid. SD, standard deviation of random residual error; Slope SD, standard deviation of random slope; top 1 min, a day-level measure defined as a value from the top 1 min out of 1440 min of a day for a given day.

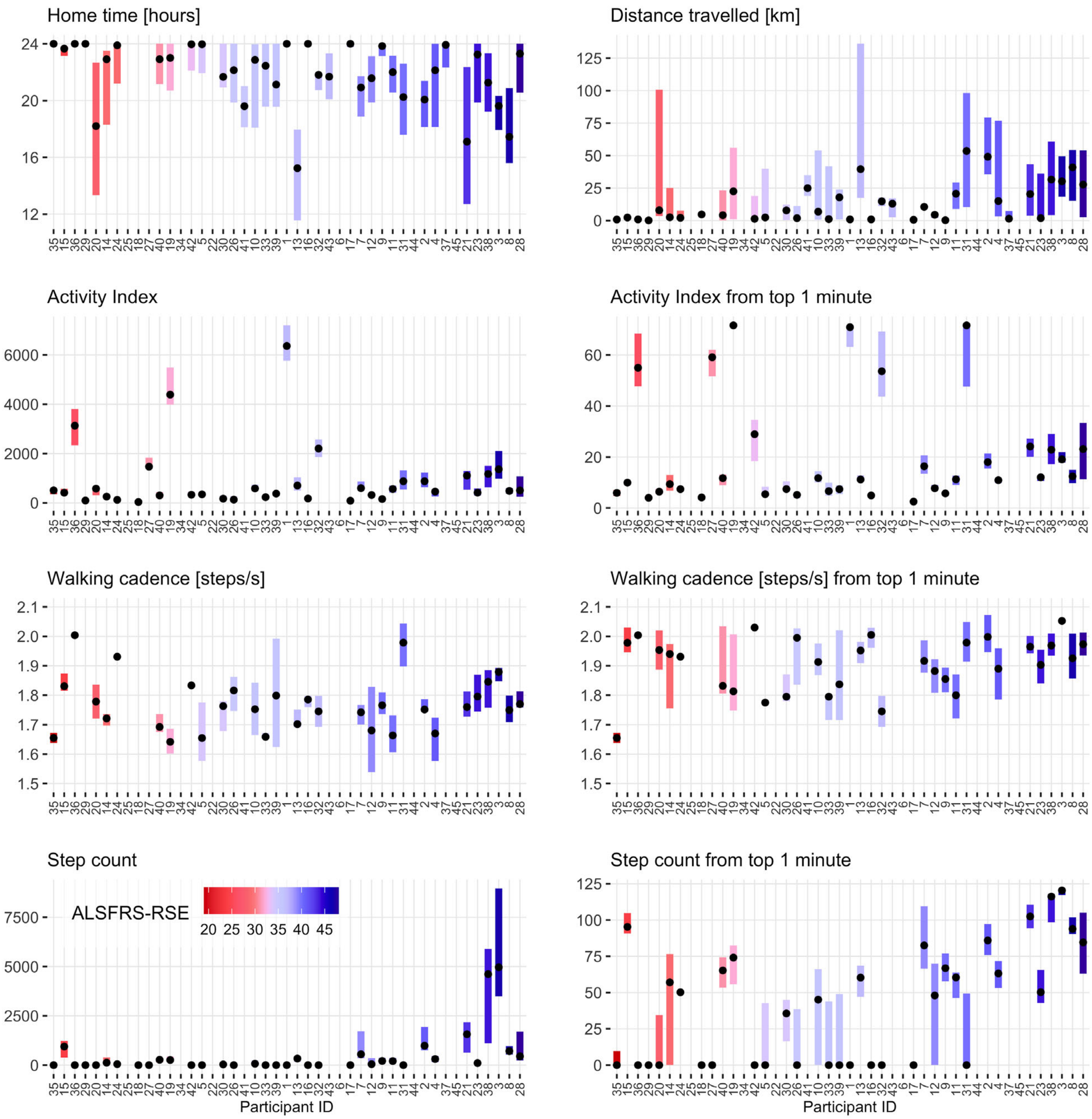

Figure 2. Median and interquartile range (IQR) of day-level measures of smartphone sensor data from 0 to 27 days after participant's first ALSFRS-RSE survey. Participants are arranged on the x-axis and the IQR bar is color-coded according to the ALSFRS-RSE total score from the first ALSFRS-RSE survey in ascending order. Day-level measures were derived from smartphone raw GPS data with imputation (home time, distance traveled) and accelerometer raw data (activity index, walking cadence, step count). For some participants, no smartphone GPS/accelerometer data were recorded in days 0–27 after the first ALSFRS-RSE survey.

When focusing on specific ALSFRS-RSE sub-scores (Table S4), we found that walking cadence from top 1 min, log(step count), and log(step count from top 1 min) remained significantly associated with ALSFRS-RSE Q4-6 (fine motor), ALSFRS-RSE Q7-9 (gross motor), and ALSFRS-RSE Q10-12 (respiratory), suggesting their consistent relevance in capturing impairments across these domains. However, for ALSFRS-RSE Q1-3 (bulbar), none of the outcome measures considered showed significant associations.

Table 4. Model intercept and estimated average outcome change associated with ALSFRS-RSE total score 1-point increase for day-level outcomes derived from smartphone sensor data. The outcome was defined as the average of day-level measures within a 21-day window centered around the ALSFRS-RSE survey response.

No.	Outcome (y)	Model intercept est. [95% CI]	Average outcome change associated with ALSFRS-RSE total score increase by 1 point – est. [95% CI] (*p*-val.)	Resid. SD	Slope SD	R2c
1	Home time (h)	23.83 [21.66, 26.00]	−0.076 [−0.136, −0.015] (*0.015)	1.274	0.010	0.677
2	log(Distance traveled, km)	0.396 [−0.492, 1.284]	0.046 [0.022, 0.071] (*<0.001)	0.488	0.004	0.756
3	log(Activity Index)	5.747 [4.895, 6.598]	0.012 [−0.009, 0.034] (0.264)	0.338	0.007	0.903
4	log(Activity Index from top 1 min)	2.263 [1.634, 2.891]	0.010 [−0.005, 0.024] (0.183)	0.193	0.018	0.940
5	Walking cadence (steps/s)	1.642 [1.537, 1.747]	0.003 [0.000, 0.006] (*0.030)	0.064	0.001	0.517
6	Walking cadence (steps/s) from top 1 min	1.642 [1.513, 1.771]	0.007 [0.003, 0.010] (*0.001)	0.076	0.005	0.472
7	log(Step count)	−1.143 [−3.559, 1.273]	0.126 [0.054, 0.197] (*0.002)	1.013	0.095	0.823
8	log(Step count from top 1 min)	−0.563 [−2.056, 0.929]	0.085 [0.044, 0.126] (*<0.001)	0.770	0.005	0.772

CI, confidence interval; Est., model estimate; *p*-val., *p*-value associated with a coefficient estimate (an asterisk denotes a BH-adjusted *p*-value being less than 0.2); R2c, LMM conditional coefficient of determination; resid. SD, standard deviation of random residual error; Slope SD, standard deviation of random slope; top 1 min, a day-level measure defined as a value from the top 1 min out of 1440 min of a day for a given day.

Discussion

Principal results

A sizeable cohort of ambulatory adults with ALS (n = 45) was enrolled and followed for 1 year. Smartphone passive and active data were collected through the Beiwe platform. The findings of this study demonstrate high feasibility and utility of remote monitoring through smartphone apps for both passive and active data collection in PALS. Smartphone accelerometer-based day-level data were used to calculate measures of walking and Activity Index, and smartphone GPS data were used to calculate day-level measures of broad mobility. Overall, ALSFRS-RSE total score and ALSFRS-RSE subscores (Q1–3, Q4–6, Q7–9, Q10–12) declined significantly over time. Of the 8 day-level measures, 4 demonstrated statistically significant monthly change: walking cadence from top 1 min, log(Activity Index from top 1 min), log(step count), and log(step count from top 1 min), and 6 were associated with the ALSFRS-RSE total score—negatively: time spent at home; positively: log(distance traveled), walking cadence, walking cadence from top 1 min, log(step count), log(step count from top 1 min). Walking cadence from the top 1 min, log(step count), and log(step count from top 1 min) were consistently associated with ALSFRS-RSE Q4-6 (fine motor), Q7-9 (gross motor), and Q10-12 (respiratory) subscores, while none of the measures showed significant associations with ALSFRS-RSE Q1-3 (bulbar), implying, not surprisingly, that the bulbar deficits might not be captured by those measures.

In our study, we observed substantial differences among participants regarding their monthly progression in ALSFRS-RSE total scores and its subscores, as well as in smartphone-based metrics of physical activity and mobility. This variability highlights the distinct progression rates of participants in various functional subdomains. We also observed variability across participants in the linear relationship between ALSFRS-RSE scores and the proposed measures. While evaluating these results, it is important to emphasize that our main objective was to demonstrate that smartphone-based digital outcomes are compatible with the current gold standard (ALSFRS-R) and to introduce interpretable and meaningful metrics that supplement the ALSFRS-R assessment. This validation against the current standard is important and necessary; our future work on smartphone-based digital metrics will further focus on “adding something” that surveys cannot capture.

Considerations for smartphone data collection

The benefits and limitations of remote digital data collection have been described previously.[14] In this study, remote data collection via Beiwe enabled physical activity monitoring in an unobtrusive manner over the course of 1 year. Participants downloaded the app, the front-end component of the platform, on their personal smartphones, avoiding the burden associated with having to wear, or carry, an additional device. Remote collection facilitated frequent sampling of ALSFRS-RSE with ease, and increased outcome measure sampling frequency may yield better accuracy in estimating average change over time, and hence decrease the trial duration and/or reduce sample size needed to observe a statistically significant effect. Furthermore, passively collected accelerometer and GPS data enable quantification of meaningful, interpretable measures on physical activity and mobility without

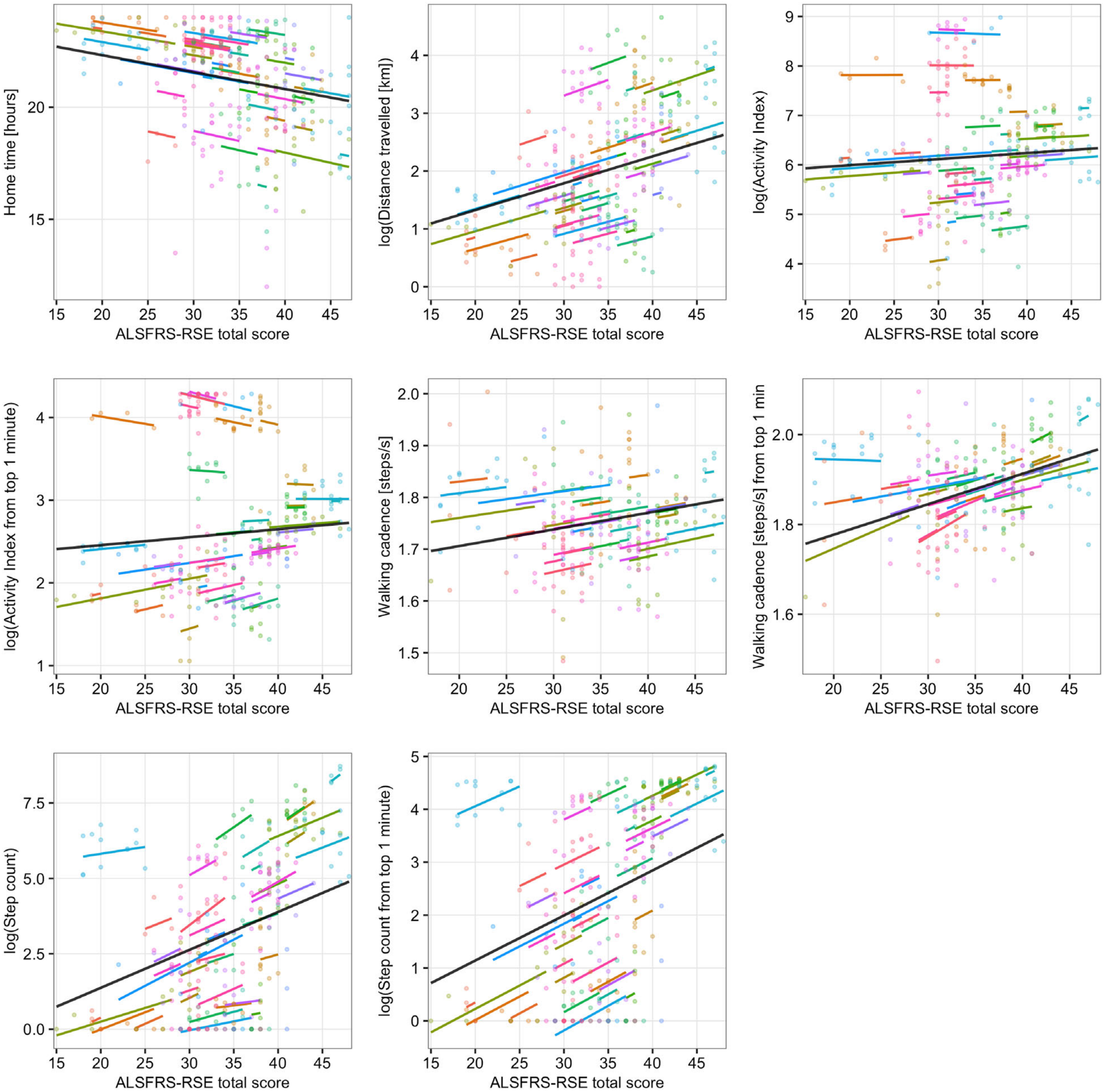

Figure 3. Association between ALSFRS-RSE total score (x-axis) and day-level outcomes based on smartphone sensor data (y-axis) estimated with linear mixed-effects models (LMMs). The outcome was defined as an average of day-level smartphone passive data measures ±10 days from the day of the ALSFRS-RSE survey response (a 21-day period in total). In each plot, points represent observed values, lines represent participants' conditional means, and both points and lines have been color-coded by participant (the colors are fixed across the panels). Black lines represent population means.

participant burden. These measures could serve as exploratory endpoints in ALSFRS-R interventional trials. This could help build evidence that the drug improves patient quality of life. Moreover, these measures could serve as primary or secondary endpoints in trials testing other interventions within the ALS community, such as those aimed at encouraging regular outdoor activities. Ultimately, this could offer a broader, more patient-centric view of treatment efficacy and quality of life.

Given the nearly ubiquitous nature of smartphones, using smartphones for digital data collection may enable greater inclusivity in clinical trials compared to wearables. While wearable devices collect biometric data, they can be prohibitively expensive, and there is a lack of diversity in studies that use wearables to study health outcomes.[36] Though smartphone ownership has skyrocketed across individuals of all income levels and races/ethnicities, ownership of additional digital devices, including wearables, has not.[37,38] Commonly cited barriers are high cost and lack of awareness and knowledge about available devices.[38] Thus, there is a nonnegligible risk of increasing existing health disparities through research using wearables for data collection.[38] Given the nearly ubiquitous nature of smartphones, using smartphones for digital data collection may enable greater inclusivity in clinical trials and even mitigate existing gaps in clinical trial diversity. Finally, the metrics investigated in this paper could potentially be used in clinical practice as digital monitoring biomarkers.

Limitations

Despite the insights gained, our study has certain limitations. First, we lacked a well-established method to account for a varying number of valid accelerometer data minutes per day. To overcome this, we constructed an analytic sample ensuring a comparable number of valid accelerometer data minutes for each participant over time. Additionally, we conducted a post-hoc analysis quantifying daily valid accelerometer minutes and daily "phone on person" minutes over time and found monthly change estimates of relatively very small magnitude that were not statistically significant.

Another limitation of our study is related to the assumption that tracking the activity recorded with a smartphone sensor equates to accurately tracking the person's overall physical activity. We acknowledge that some individuals may not always carry their smartphones with them, which can lead to gaps in monitoring. This might be particularly problematic for individuals who engage in indoor walking or physical activity, during which their smartphones might be set aside, resulting in an absence of monitoring during those periods. While using smartphone accelerometer data-derived measures may not be the most optimal method for measuring the total volume of physical activity, we consider it a justifiable approach for assessing peak performance, such as tracking step count from top 1 min, or measuring temporal indicators, such as walking cadence, especially given our study's extended 1-year duration.

Finally, we recognize that smartphone wear and usage variations could affect our measures in ALS patients with poor hand function. Therefore, we emphasized data streams and measures that are robust to variations in smartphone wear. GPS-based measures are not affected either by hand function or phone wear characteristics, and the algorithm we used to process accelerometer-based walking measures is invariant to phone orientation and translation and has been validated across different sensor locations.[39,40] However, the scope of robust and easy-to-understand measures is limited. Therefore, further research into interpretable metrics derived from passively collected smartphone accelerometer data, which could provide an unambiguous interpretation of ALS progression, is warranted.

Conclusions

In this study, smartphones were used to collect both passive (accelerometer, GPS) and active (ALSFRS-RSE survey) data over the course of 1 year in a cohort of PALS. Smartphone accelerometer and GPS sensor data were used to derive day-level measures of walking, activity index, and mobility. The ALSFRS-RSE was administered through the Beiwe app. We demonstrated that out of eight day-level measures, four showed a significant decline over time and six were significantly associated with ALSFRS-RSE total score. Smartphones provide a unique opportunity for monitoring patients in natural settings both clinical trials and clinical practice, and the prevalence of the devices may help reduce existing health and research disparities. Looking ahead, greater adoption of digital outcome measures in interventional clinical trials is needed to elucidate their potential to enhance clinical trial efficiency and serve as primary outcome measures.

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Author Contributions

Concept and design—M.K., S.A.J., M.S., K.M.B., S.I., A.L., J.D.B., J.P.O. Data collection—S.A.J., Z.A.S., A.P.C., A.S.I. Statistical analysis—M.K. Interpretation of results—M.K., J.O., J.P.O., J.D.B. Manuscript preparation—M.K., J.O. Critical review of manuscript—M.K., J.O.,

S.A.J., M.S., K.M.B., S.I., A.L., Z.A.S., A.P.C., A.S.I., E.H., J.D.B., J.P.O. Study supervision—J.P.O., J.D.B.

Conflict of Interest

Marta Karas, Julia Olsen, Marcin Straczkiewicz, Katherine M. Burke, Zoe A. Scheier, Alison P. Clark, Amrita S. Iyer, Emily Huang, Jukka-Pekka Onnela—report no competing interests.

Stephen A. Johnson—reports research support from the ALS Association.

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Amir Lahav—paid strategic advisor for digital health at Mitsubishi Tanabe Pharma America, Inc. He has been paid for consulting services for Pfizer, Redenlab, Curebase, Castor, Lapsi Health, Solo AI, Quantificare, Modality AI, Videra Health, Neuralight, and the Global Platform Alzheimer's Foundation.

James D. Berry—reports research support from Biogen, MT Pharma Holdings of America, Transposon Therapeutics, Alexion, Rapa Therapeutics, ALS Association, Muscular Dystrophy Association, ALS One, Tambourine, ALS Finding a Cure. He has been a paid member of an advisory panel for Regeneron, Biogen, Clene Nanomedicine, Mitsubishi Tanabe Pharma Holdings America, Inc., Janssen, RRT. He received an honorarium for educational events for Projects in Knowledge and the Muscular Dystrophy Association. He has unpaid roles on the advisory boards for the nonprofits Everything ALS and ALS One.

Declaration of Generative AI and AI-Assisted Technologies in the Writing Process

During the preparation of this work, the authors used ChatGPT in order to improve the language and readability of the written text. After using this tool/service, the authors reviewed and edited the content as needed and took full responsibility for the content of the publication.

Data Availability Statement

The de-identified data are available upon reasonable request. R code for all data preprocessing and data analysis is publicly available on GitHub repository (https://github.com/onnela-lab/als-beiwe-passive-data).

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Supporting Information

Additional supporting information may be found online in the Supporting Information section at the end of the article.

Data S1.

